INCH POUND

MIL-P-71095 (AR)
15 September 1993

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE: M864
WEATHERSEAL ASSEMBLY FOR

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality provisions, and packaging for the weatherseal assembly for use with projectile, 155MM, HE: M864.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-T-680 - Tape Pressure Sensitive Adhesive, Packaging and Packing of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications for a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.9)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (AREC)

PRODUCT DRAWING

9390991 - Weatherseal Assembly

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3330	-	Peel Adhesion of Pressure-Sensitive Tape at 180 degrees Angle
ASTM D3652	-	Thickness of Pressure-Sensitive and Gummed Tapes
ASTM D3654	-	Holding Power of Pressure-Sensitive Tapes
ASTM D3715	-	Quality Assurance of Pressure-Sensitive Tapes
ASTM D3759	-	Tensile Strength and Elongation of Pressure-Sensitive Tapes
ASTM D3815	-	Accelerated Aging of Pressure Sensitive Tapes by Carbon-Arc Exposure Apparatus
ASTM D3833	-	Water Vapor Transmission of Pressure-Sensitive Tapes

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.1.1 Adhesive. The adhesive shall be pressure sensitive acrylic and require no solvent, heat, or other preparation prior to or after application to clean, dry surfaces. The adhesive shall be homogeneous, water resistant, and applied in a smooth uniform layer to the aluminum foil side of a polyvinyl fluoride/aluminum foil backing (dwg. 9390991).

3.1.2 Release liner. The weatherseal assembly shall be adhered to a release liner made of a smooth paper backing.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9390991, all associated drawings, and with all requirements specified in applicable specifications and drawings.

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3.3 Physical properties. The weatherseal assembly material (dwg. 9390991) shall comply with the requirements of Table I when tested as specified in 4.4.3.

3.3.1 Resistance to weathering. When tested as specified in 4.4.3.4, the weatherseal assembly material shall show no visual evidence of lifting, curling, buckling, component separation, or other degradation that would make it unfit for an external sealing media.

3.3.2 Corrosivity. Following accelerated aging (see 4.4.3.4), the test panels shall show no evidence of pitting or corrosion in the areas covered by the weatherseal assembly material specimens.

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TABLE I. - Physical properties.

Property	Requirements Minimum	Requirements Maximum	Inspection Method Reference
Thickness, Backing (inches)	0.0095	0.0125	4.4.3.1
Thickness, Adhesive (inches)	0.0015	0.0020	4.4.3.1
Tensile Strength (lbs/inch width)	100	-	4.4.3.2
Adhesion (ounces/inch width): Initial	55	-	4.4.3.3
After accelerated aging	50	-	4.4.3.3
Adhesive Transfer (percent): Initial	-	5	4.4.3.3
After accelerated aging	-	10	4.4.3.3
Water Vapor Transmission Rate (grams/100 sq. inches/ 24 hrs)	-	0.03	ASTM D 3833
Resistance to Weathering	See 3.3.1	-	4.4.3.4
Corrosivity	-	See 3.3.2	4.4.3.4
Static Load (hours)	30	-	ASTM D 3654
Adhesion between polyvinyl fluoride and aluminum foil (ounces/inch width) Initial	48	-	ASTM D 3330
After accelerated aging	70	-	ASTM D 3330

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3.4 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.5 Suitability for use with explosive. When so specified, application shall be made to a government laboratory or other government approved laboratory designated by the procuring agency for determination of the ability of the tape for use with a particular explosive or explosives.

3.6 Workmanship. Workmanship shall be in accordance with the best commercial practice. The weatherseal assembly shall be clean and free from foreign matter and defects that impair its serviceability or appearance.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and test) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance or defective material.

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4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

PART DESCRIPTION	DRAWING	QUANTITY
Weatherseal assemblies	9390991	30
Weatherseal assembly material thickness, test specimens	-	30
Tensile test specimens	-	30
Adhesion test specimens	-	60
Water penetration test specimens	-	30
Corrosivity and resistance to weathering test specimens	-	30
Static load test specimens	-	30
Adhesion between polyvinyl fluoride and aluminum test specimens	-	60

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4.3.1 Assemblies and components. All assemblies and components shall have been produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of ninety days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see Table II) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE II. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-P-71095 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2	DRAWING NUMBER
	Weatherseal Assembly			9390991
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY: None / INSPECTION METHOD REFERENCE
	Weatherseal Assembly (Dwg. 9390991)			
	Diameter	30,100%	3.2	Gage
	Thickness of Adhesive	30,100%	3.3	4.4.3.1
	Thickness of backing	30,100%	3.3	4.4.3.1
	Tensile test	30,100%	3.3	4.4.3.2
	Adhesion, initial	30,100%	3.3	4.4.3.3
	Adhesion, after accelerated aging	30,100%	3.3	4.4.3.3
	Adhesive transfer, initial	30,100%	3.3	4.4.3.3
	Adhesive transfer, after accelerated aging	30,100%	3.3	4.4.3.3
	Water vapor transmission	30,100%	3.3	ASTM D 3833
	Resistance to weathering	30,100%	3.3	4.4.3.4
	Corrosivity	30,100%	3.3	4.4.3.4
	Static load test	30,100%	3.3	ASTM D 3654 Procedure C

NOTES:

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TABLE II. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-P-71095 (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
	Weatherseal Assembly		9390991
			NEXT HIGHER ASSEMBLY: None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Adhesion between polyvinyl fluoride and aluminum, initial	30,100%	3.3	ASTM D 3330
	Adhesion between polyvinyl fluoride and aluminum, after accelerated aging	30,100%	3.3	ASTM D 3330
	Evidence of poor workmanship	30,100%	3.5	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with the applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials, the same material batch, and the same methods, in accordance with the same drawing, same drawing revision, same specification revision and manufactured under essentially the same conditions, and at essentially the same time. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.4 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with applicable drawi minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table III, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE III Attributes sampling inspection

Lot Size	Inspection Levels I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 TO 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicates one hundred percent inspection. If sample size exceeds lot size, pe hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the government for evaluation (see 6.5). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternate provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

c. Examination. See MIL-A-48078.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.1	Weatherseal Assembly	SHEET 1 OF 2	9390991
			NEXT HIGHER ASSEMBLY: None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter	Level V	3.2	Gage
102	Thickness of adhesive	Note a	3.3	4.4.3.1
103	Thickness of backing	Note a	3.3	4.4.3.1
104	Tensile test	Note a	3.3	4.4.3.2
105	Adhesion, initial	Note a	3.3	4.4.3.3
106	Adhesion, after accelerated aging	Note a	3.3	4.4.3.3
107	Adhesive transfer, initial	Note a	3.3	4.4.3.3
108	Adhesive transfer, after accelerated aging	Note a	3.3	4.4.3.3
109	Water vapor transmission	Note a	3.3	ASTM D3833
110	Resistance to weathering	Note a	3.3	4.4.3.4
111	Corrosivity	Note a	3.3	4.4.3.4
112	Static load test	Note a	3.3	ASTM D3654 Procedure C

NOTES:

a. Samples, shall be selected in accordance with ASTM D3715. The sample size shall be as for a normal inspection by attributes, however, the quantity represented by the sample shall be accepted on zero defects, and rejected on one or more defects when the samples are tested in accordance with each applicable test method.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Weatherseal Assembly	2 OF 2	9390991
			NEXT HIGHER ASSEMBLY: None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Adhesion between polyvinyl fluoride and aluminum, initial	Note a	3.3	ASTM D 3330
202	Adhesion between polyvinyl fluoride and aluminum, after accelerated aging	Note a	3.3	ASTM D 3330
203	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

a. Samples shall be selected in accordance with ASTM D 3715. The sample size shall be as for a normal inspection by attributes, however, the quantity represented by the sample shall be accepted on zero defects, and rejected on one or more defects when the samples are tested in accordance with each applicable test method.

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4.4.3 Methods of inspection.

4.4.3.1 Thickness of weatherseal backing and adhesive. The thickness of the weatherseal assembly material shall be determined in accordance with ASTM D 3652. When the total thickness is determined, the adhesive shall be completely removed from the backing using a suitable sois treatment shall be performed in a manner that will leave the backing smooth for testing. The thickness of the backing material shall then be determined. The adhesive thickness shall be the backing thickness subtracted from the total thickness.

4.4.3.2 Tensile test. The machine direction tensile strength of the weatherseal assembly material shall be determined in accordance with ASTM D 3759 except that the apparatus need not be capable of measuring elongation.

4.4.3.3 Adhesion and adhesive transfer. The adhesion of the weatherseal assembly material, both as received and after the specimens have been subjected to accelerated aging in accordance with ASTM D 3815, shall be determined in accordance with ASTM D 3330 Method A. After removal of each specimen from the test panel, the panel shall be visually examined for adhesive transfer.

4.4.3.4 Resistance to weathering and corrosivity. Weatherseal assembly materiel specimens shall undergo accelerated aging in accordance with ASTM D 3815. Each sample shall be visually inspected for resistance to weathering. After removal of each specimen from the test panel, the test panel shall be visually inspected for corrosion in the areas that were covered by the test samples.

4.5 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attribute data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

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5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified in accordance with the applicable requirements of PPP-T-680.

5.2 Packing. Packing shall be level A, B, or C as specified in accordance with the applicable requirements of PPP-T-680.

5.3 Marking. Containers of the weatherseal shall be marked in accordance with the applicable requirements of PPP-T-680.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This weatherseal is intended for use on the Projectile, 155MM, HE: M864.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).

c. Requirements for submission of first article sample.

d. Certificate of conformance for each lot of shipment of product.

6.3 Submission of contractor inspection design for approval. Submit copiired to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAR-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Inspection lots. It is noted that the size of the inspection lots of components and assemblies may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria per MIL-STD-1168 and sampling provisions of this specification are maintained.

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6.5 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.5.1 Visual examionation qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.6 Visual standards. Visual standards shall be established at, or provided to, the contractor's plant in accordance with 3.5.

6.7 Submission of results of contractor-conducted examinations and tests. Data shall be submitted in accordance with item on DD Form 1423 for the contract.

6.8 Submission of ammunition data cards. In addition to the normal distribution of records, one (1) copy of all ammunition data cards shall be forwarded to Commander, ARDEC, Picatinny Arsenal, NJ 07086-5000, ATTN: SMCAR-QAR-Q.

6.9 Drawings. Drawings listed in section 2 of this specification under the heading U.S. Army Armament, Research and Development Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinnt Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.10 Subject term (key word) listing.

Seal
Aluminum foil
Adhesive
Pressure-sensitive

Custodian:
Army-AR

Preparing activity:
Army-AR
(Project 1320-AC75)

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